

PRACTICAL - 1

show databases;

	Database
▶	information_schema
	mysql
	performance_schema
	startersql
	student
	sys

create database employee;

use employee;

create table emp_details(emp_no int(10),emp_name varchar(30),emp_gender
varchar(1),emp_sal int(30));

show tables;

	Tables_in_employee
▶	emp_details

alter table emp_details add emp_dept varchar(20);

desc emp_details;

	Field	Type	Null	Key	Default	Extra
▶	emp_no	int	YES		NULL	
	emp_name	varchar(30)	YES		NULL	
	emp_gender	varchar(1)	YES		NULL	
	emp_sal	int	YES		NULL	
	emp_dept	varchar(20)	YES		NULL	

insert into emp_details values(1,'Ram','M',300000,'designing');

insert into emp_details values(2,'Soham','M',300000,'designing');

```
insert into emp_details values(3,'Mohan','M',250000,'management');
```

```
insert into emp_details values(4,'Om','M',400000,'coding');
```

```
select * from emp_details;
```

	emp_no	emp_name	emp_gender	emp_sal	emp_dept
▶	1	Ram	M	300000	designing
	2	Soham	M	300000	designing
	3	Mohan	M	250000	management
	4	Om	M	400000	coding

```
create table emp_info as select emp_no,emp_name,emp_gender from emp_details;
```

```
select * from emp_info;
```

	emp_no	emp_name	emp_gender
▶	1	Ram	M
	2	Soham	M
	3	Mohan	M
	4	Om	M

```
truncate table emp_info;
```

```
select * from emp_info;
```

	emp_no	emp_name	emp_gender

```
SET SQL_SAFE_UPDATES = 0;
```

```
drop table emp_info;
```

```
create view emp_view1 as select * from emp_details;
```

```
create view emp_view2 as select * from emp_details where emp_dept="designing";
```

```
select * from emp_view1;
```

	emp_no	emp_name	emp_gender	emp_sal	emp_dept
▶	1	Ram	M	300000	designing
	2	Soham	M	300000	designing
	3	Mohan	M	250000	management
	4	Om	M	400000	coding

select * from emp_view2;

	emp_no	emp_name	emp_gender	emp_sal	emp_dept
▶	1	Ram	M	300000	designing
	2	Soham	M	300000	designing

update emp_details set emp_dept="coding" where emp_name="Mohan";

select * from emp_details;

	emp_no	emp_name	emp_gender	emp_sal	emp_dept
▶	1	Ram	M	300000	designing
	2	Soham	M	300000	designing
	3	Mohan	M	250000	coding
	4	Om	M	400000	coding

drop view emp_view1;

drop view emp_view2;

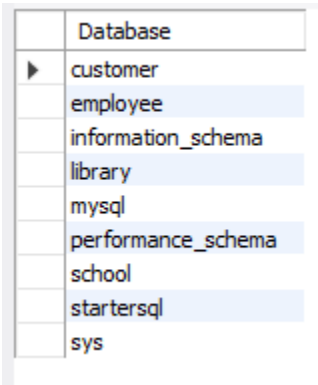
create index emp_ind on emp_details(emp_no,emp_name);

show index from emp_details;

	Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
▶	emp_details	1	emp_ind	1	emp_no	A	4	NULL	NULL	YES	BTREE			YES	NULL
	emp_details	1	emp_ind	2	emp_name	A	4	NULL	NULL	YES	BTREE			YES	NULL

PRACTICAL - 2

show databases;

A screenshot of a MySQL database interface showing a list of databases. The 'Database' column lists: customer, employee, information_schema, library, mysql, performance_schema, school, startersql, and sys. The 'employee' and 'performance_schema' databases are highlighted in blue.

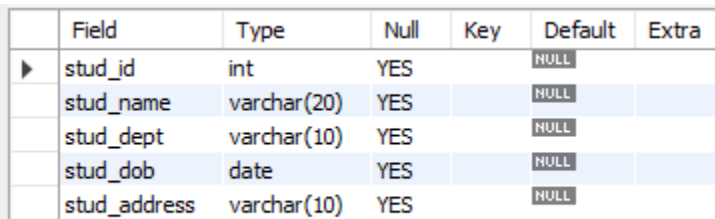
	Database
▶	customer
	employee
	information_schema
	library
	mysql
	performance_schema
	school
	startersql
	sys

create database student;

use student;

create table stud_tab(stud_id int(4),stud_name varchar(20),stud_dept varchar(10),stud_dob date,stud_address varchar(10));

desc stud_tab;

A screenshot of a MySQL table structure view for the 'stud_tab' table. It shows columns: stud_id (int), stud_name (varchar(20)), stud_dept (varchar(10)), stud_dob (date), and stud_address (varchar(10)). All columns are nullable (YES) and have a default value of NULL.

	Field	Type	Null	Key	Default	Extra
▶	stud_id	int	YES		NULL	
	stud_name	varchar(20)	YES		NULL	
	stud_dept	varchar(10)	YES		NULL	
	stud_dob	date	YES		NULL	
	stud_address	varchar(10)	YES		NULL	

insert into stud_tab values(1,'Ram','Comp','2002-11-05','Pune');

insert into stud_tab values(2,'Soham','IT','2002-09-03','Nashik');

insert into stud_tab values(3,'Ramesh','Comp','2002-03-19','Pune');

insert into stud_tab values(4,'Mohan','AI&DS','2002-02-22','Nagpur');

select * from stud_tab;

	stud_id	stud_name	stud_dept	stud_dob	stud_address
▶	1	Ram	Comp	2002-11-05	Pune
	2	Soham	IT	2002-09-03	Nashik
	3	Ramesh	Comp	2002-03-19	Pune
	4	Mohan	AI&DS	2002-02-22	Nagpur

```
alter table stud_tab add shift varchar(10);
```

```
update stud_tab set shift='first' where stud_id=1;
```

```
update stud_tab set shift='second' where stud_id=2;
```

```
update stud_tab set shift='first' where stud_id=3;
```

```
update stud_tab set shift='first' where stud_id=4;
```

```
select * from stud_tab;
```

	stud_id	stud_name	stud_dept	stud_dob	stud_address	shift
▶	1	Ram	Comp	2002-11-05	Pune	first
	2	Soham	IT	2002-09-03	Nashik	second
	3	Ramesh	Comp	2002-03-19	Pune	first
	4	Mohan	AI&DS	2002-02-22	Nagpur	first

```
insert into stud_tab values(5,'Omkar','ENTC','2002-06-26','Pune','second');
```

```
select * from stud_tab;
```

	stud_id	stud_name	stud_dept	stud_dob	stud_address	shift
▶	1	Ram	Comp	2002-11-05	Pune	first
	2	Soham	IT	2002-09-03	Nashik	second
	3	Ramesh	Comp	2002-03-19	Pune	first
	4	Mohan	AI&DS	2002-02-22	Nagpur	first
	5	Omkar	ENTC	2002-06-26	Pune	second

```
delete from stud_tab where stud_address='Nagpur';
```

```
select * from stud_tab;
```

	stud_id	stud_name	stud_dept	stud_dob	stud_address	shift
▶	1	Ram	Comp	2002-11-05	Pune	first
	2	Soham	IT	2002-09-03	Nashik	second
	3	Ramesh	Comp	2002-03-19	Pune	first
	5	Omkar	ENTC	2002-06-26	Pune	second

update stud_tab set stud_id=4 where stud_name='Omkar';

select * from stud_tab;

	stud_id	stud_name	stud_dept	stud_dob	stud_address	shift
▶	1	Ram	Comp	2002-11-05	Pune	first
	2	Soham	IT	2002-09-03	Nashik	second
	3	Ramesh	Comp	2002-03-19	Pune	first
	4	Omkar	ENTC	2002-06-26	Pune	second

select * from stud_tab where stud_dob between '2002-01-01' and '2002-07-01';

	stud_id	stud_name	stud_dept	stud_dob	stud_address	shift
▶	3	Ramesh	Comp	2002-03-19	Pune	first
	4	Omkar	ENTC	2002-06-26	Pune	second

alter table stud_tab add stud_fees int(15);

update stud_tab set stud_fees=15000 where stud_id=1;

update stud_tab set stud_fees=20000 where stud_id=2;

update stud_tab set stud_fees=20000 where stud_id=3;

update stud_tab set stud_fees=15000 where stud_id=4;

select * from stud_tab;

	stud_id	stud_name	stud_dept	stud_dob	stud_address	shift	stud_fees
▶	1	Ram	Comp	2002-11-05	Pune	first	15000
	2	Soham	IT	2002-09-03	Nashik	second	20000
	3	Ramesh	Comp	2002-03-19	Pune	first	20000
	4	Omkar	ENTC	2002-06-26	Pune	second	15000

```
select * from stud_tab where stud_fees=(select max(stud_fees) from stud_tab);
```

	stud_id	stud_name	stud_dept	stud_dob	stud_address	shift	stud_fees
▶	2	Soham	IT	2002-09-03	Nashik	second	20000
	3	Ramesh	Comp	2002-03-19	Pune	first	20000

```
select sum(stud_fees) from stud_tab;
```

	sum(stud_fees)
▶	70000

```
create table stud_info as select stud_id,stud_name from stud_tab;
```

```
select stud_id from stud_tab union select stud_id from stud_info;
```

	stud_id
▶	1
	2
	3
	4

PRACTICAL - 3

show databases;

	Database
▶	information_schema
	library
	mysql
	performance_schema
	school
	startersql
	student
	sys

create database customer;

show databases;

	Database
▶	customer
	information_schema
	library
	mysql
	performance_schema
	school
	startersql
	student
	sys

use customer;

create table cust_tab(id int(4),name varchar(20),quantity int(4),price int(10),item varchar(20));

show tables;

	Tables_in_customer
▶	cust_tab

create table cust_info(id int(4),name varchar(20),address varchar(20),mobile varchar(20));


```
desc cust_tab;
```

	Field	Type	Null	Key	Default	Extra
►	id	int	YES		NULL	
	name	varchar(20)	YES		NULL	
	quantity	int	YES		NULL	
	price	int	YES		NULL	
	item	varchar(20)	YES		NULL	

```
desc cust_info;
```

	Field	Type	Null	Key	Default	Extra
►	id	int	YES		NULL	
	name	varchar(20)	YES		NULL	
	address	varchar(20)	YES		NULL	
	mobile	varchar(20)	YES		NULL	

```
alter table cust_tab add primary key(id);
```

```
alter table cust_info add foreign key(id) references cust_tab(id);
```

```
insert into cust_tab values(1,'Ram',1,15,'Milk');
```

```
insert into cust_tab values(2,'Soham',2,20,'Toast');
```

```
insert into cust_tab values(3,'Mohan',4,5,'Parle-G');
```

```
insert into cust_tab values(4,'Om',2,20,'Coca Cola');
```

```
select * from cust_tab;
```

	id	name	quantity	price	item
►	1	Ram	1	15	Milk
	2	Soham	2	20	Toast
	3	Mohan	4	5	Parle-G
	4	Om	2	20	Coca Cola
*	NULL	NULL	NULL	NULL	NULL

```
alter table cust_tab add totalprice int(4);
```

```
update cust_tab set totalprice=quantity*price where id=1;
```

```
update cust_tab set totalprice=quantity*price where id=2;
```

```
update cust_tab set totalprice=quantity*price where id=3;
```

```
update cust_tab set totalprice=quantity*price where id=4;
```

```
select * from cust_tab;
```

	id	name	quantity	price	item	totalprice
▶	1	Ram	1	15	Milk	15
	2	Soham	2	20	Toast	40
	3	Mohan	4	5	Parle-G	20
	4	Om	2	20	Coca Cola	40
*	NULL	NULL	NULL	NULL	NULL	NULL

```
insert into cust_info values(1,'Ram','Pune','9943569081');
```

```
insert into cust_info values(2,'Soham','Pune','9978491281');
```

```
insert into cust_info values(3,'Mohan','Nashik','8782356712');
```

```
insert into cust_info values(4,'Om','Nagpur','7823450189');
```

```
select * from cust_info;
```

	id	name	address	mobile
▶	1	Ram	Pune	9943569081
	2	Soham	Pune	9978491281
	3	Mohan	Nashik	8782356712
	4	Om	Nagpur	7823450189

```
select cust_tab.id,cust_tab.name,cust_tab.item,cust_info.address from cust_tab inner join  
cust_info on cust_tab.id=cust_info.id;
```

	id	name	item	address
▶	1	Ram	Milk	Pune
	2	Soham	Toast	Pune
	3	Mohan	Parle-G	Nashik
	4	Om	Coca Cola	Nagpur

```
select cust_tab.id,cust_tab.name,cust_tab.item,cust_info.address from cust_tab left outer join  
cust_info on cust_tab.id=cust_info.id;
```

	id	name	item	address
▶	1	Ram	Milk	Pune
	2	Soham	Toast	Pune
	3	Mohan	Parle-G	Nashik
	4	Om	Coca Cola	Nagpur

```
select cust_tab.id,cust_tab.name,cust_tab.item,cust_info.address from cust_tab right outer  
join cust_info on cust_tab.id=cust_info.id;
```

	id	name	item	address
▶	1	Ram	Milk	Pune
	2	Soham	Toast	Pune
	3	Mohan	Parle-G	Nashik
	4	Om	Coca Cola	Nagpur

```
create view mul_view as select cust_tab.id,cust_tab.name,cust_info.address from  
cust_tab,cust_info where cust_info.id=cust_tab.id;
```

```
select * from mul_view;
```

	id	name	address
▶	1	Ram	Pune
	2	Soham	Pune
	3	Mohan	Nashik
	4	Om	Nagpur

PRACTICAL - 4

```
create database library;
```

```
use library;
```

```
create table Borrower(Rollno int(4),Name varchar(20),DateofIssue date,NameofBook  
varchar(30),Status varchar(10));
```

```
insert into Borrower values(14,'Ram','2022-09-19','Operating System','I');
```

```
insert into Borrower values(27,'Soham','2022-07-24','Object Oriented Programming','I');
```

```
insert into Borrower values(34,'Mohan','2022-06-12','Microprocessor','I');
```

```
insert into Borrower values(48,'Om','2022-04-19','Mechanics','I');
```

```
select * from Borrower;
```

	Rollno	Name	DateofIssue	NameofBook	Status
►	14	Ram	2022-09-19	Operating System	I
	27	Soham	2022-07-24	Object Oriented Programming	I
	34	Mohan	2022-06-12	Microprocessor	I
	48	Om	2022-04-19	Mechanics	I

```
create table Fine(Rollno int(4),Date date,Amount int(10));
```

```
delimiter //
```

```
create procedure calc_Fine(in r int(10),in b varchar(30))
```

```
begin
```

```
declare doi date;
```

```
declare diff int(3);
```

```
select DateofIssue into doi from Borrower where Rollno=r and NameofBook=b;
```

```
select datediff(curdate(),doi) into diff;
```

```
if diff>=15 and diff<=30 then
```

```
insert into Fine values(r,curdate(),diff*5);
```

```
end if;
```

```
if diff>30 then
```

```
insert into Fine values(r,curdate(),diff*50);
```

```
end if;
```

```
end//
```

```
delimiter //
```

```
create procedure submit(in r int(2))
```

```
begin
```

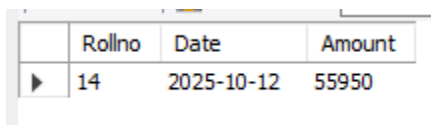
```
update Borrower set Status='R' where Rollno=r;
```

```
delete from Fine where Rollno=r;
```

```
end//
```

```
call calc_Fine(14,'Operating System');
```

```
select * from Fine;
```



	Rollno	Date	Amount
▶	14	2025-10-12	55950

```
call calc_Fine(27,'Object Oriented Programming');
```

```
call calc_Fine(34,'Microprocessor');
```

```
call calc_Fine(48,'Mechanics');
```

```
select * from Fine;
```

	Rollno	Date	Amount
▶	14	2025-10-12	55950
	27	2025-10-12	58800
	34	2025-10-12	60900
	48	2025-10-12	63600

```
call submit(14);
```

```
call submit(27);
```

```
call submit(48);
```

```
call submit(34);
```

```
select * from Fine;
```

	Rollno	Date	Amount
--	--------	------	--------

```
select * from Borrower;
```

	Rollno	Name	DateofIssue	NameofBook	Status
▶	14	Ram	2022-09-19	Operating System	R
	27	Soham	2022-07-24	Object Oriented Programming	R
	34	Mohan	2022-06-12	Microprocessor	R
	48	Om	2022-04-19	Mechanics	R

PRACTICAL - 5

-- Step 1: Create the table

```
CREATE TABLE areas (  
    radius DECIMAL(5,2),  
    area DECIMAL(10,2)  
);
```

-- Step 2: Use a stored procedure (MySQL equivalent of PL/SQL block)

```
DELIMITER $$
```

```
CREATE PROCEDURE calc_area()  
BEGIN  
    DECLARE r INT DEFAULT 5;  
    DECLARE a DECIMAL(10,2);  
  
    WHILE r <= 9 DO  
        SET a = 3.14159 * r * r;  
        INSERT INTO areas (radius, area) VALUES (r, a);  
        SET r = r + 1;  
    END WHILE;  
  
END$$
```

```
DELIMITER ;
```

-- Step 3: Execute the procedure

```
CALL calc_area();
```

-- Step 4: View the results

```
SELECT * FROM areas;
```

	radius	area
▶	5.00	78.54
	6.00	113.10
	7.00	153.94
	8.00	201.06
	9.00	254.47

PRACTICAL - 6

```
create database Score;
```

```
use Score;
```

```
create table stud_marks(name varchar(20),total_marks int(5));
```

```
create table Result(roll_no int(3) primary key,name varchar(20),class varchar(20));
```

```
insert into stud_marks values('Suresh',995);
```

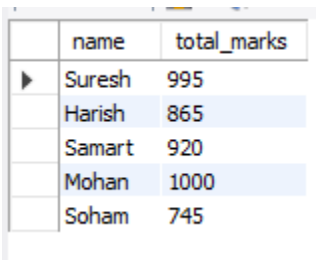
```
insert into stud_marks values('Harish',865);
```

```
insert into stud_marks values('Samart',920);
```

```
insert into stud_marks values('Mohan',1000);
```

```
insert into stud_marks values('Soham',745);
```

```
select * from stud_marks;
```



	name	total_marks
▶	Suresh	995
	Harish	865
	Samart	920
	Mohan	1000
	Soham	745

```
insert into Result(roll_no,Name) values(1,'Suresh');
```

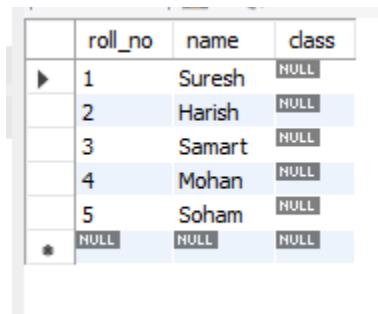
```
insert into Result(roll_no,Name) values(2,'Harish');
```

```
insert into Result(roll_no,Name) values(3,'Samart');
```

```
insert into Result(roll_no,Name) values(4,'Mohan');
```

```
insert into Result(roll_no,Name) values(5,'Soham');
```

```
select * from Result;
```



	roll_no	name	class
▶	1	Suresh	NULL
	2	Harish	NULL
	3	Samart	NULL
	4	Mohan	NULL
	5	Soham	NULL
*	NULL	NULL	NULL

```
delimiter //
```

```
create procedure proc_Grade(in r int(2),out grade char(25))
```

```
begin
```

```
declare m int(4);
```

```
select total_marks into m from stud_marks where name=(select name from Result where  
roll_no=r);
```

```
if m>=990 and m<=1500 then
```

```
select 'Distinction' into grade;
```

```
update Result set Class='Distinction' where Roll_no=r;
```

```
elseif m>=900 and m<=989 then
```

```
select 'FirstClass' into grade;
```

```
update Result set Class='FirstClass' where Roll_no=r;
```

```
elseif m>=825 and m<=899 then
```

```
select 'SecondClass' into grade;
```

```
update Result set Class='SecondClass' where Roll_no=r;
```

else

select '--' into grade;

update Result set Class='--' where Roll_no=r;

end if;

end //

delimiter //

create function func_Grade(r int(2))

returns varchar(25)

deterministic

begin

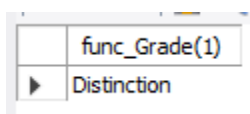
declare grade varchar(25);

call proc_Grade(r,grade);

return grade;

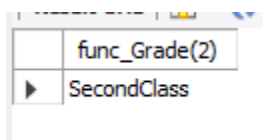
end //

select func_Grade(1); //



func_Grade(1)
Distinction

select func_Grade(2); //



func_Grade(2)
SecondClass

```
select func_Grade(3); //
```

	func_Grade(3)
▶	FirstClass

```
select func_Grade(4); //
```

	func_Grade(4)
▶	Distinction

```
select func_Grade(5); //
```

	func_Grade(5)
▶	--

```
select * from Result; //
```

	roll_no	name	class
▶	1	Suresh	Distinction
	2	Harish	SecondClass
	3	Samart	FirstClass
	4	Mohan	Distinction
	5	Soham	--
*	NULL	NULL	NULL

PRACTICAL - 7

```
create database class;
```

```
use class;
```

```
create table O_RollCall(roll_no int(3),name varchar(20));
```

```
create table N_RollCall(roll_no int(3),name varchar(20));
```

```
insert into O_RollCall values (1,'Himanshu');
```

```
insert into O_RollCall values (2,'Ram');
```

```
insert into O_RollCall values (3,'Soham');
```

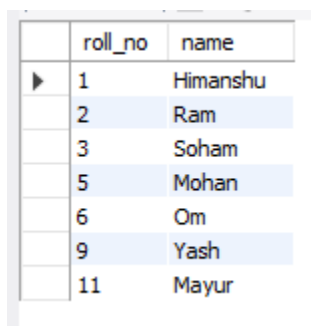
```
insert into O_RollCall values (5,'Mohan');
```

```
insert into O_RollCall values (6,'Om');
```

```
insert into O_RollCall values (9,'Yash');
```

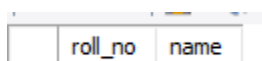
```
insert into O_RollCall values (11,'Mayur');
```

```
select * from O_RollCall;
```



	roll_no	name
▶	1	Himanshu
	2	Ram
	3	Soham
	5	Mohan
	6	Om
	9	Yash
	11	Mayur

```
select * from N_RollCall;
```



	roll_no	name
--	---------	------

```
delimiter //
```

```
create procedure cursor_proc_p1()

begin

declare fin integer default 0;

declare old_roll int(3);

declare old_name varchar(20);

declare new_roll int(3);

declare old_csr cursor for select roll_no,name from O_RollCall;

declare new_csr cursor for select roll_no from N_RollCall;

declare continue handler for not found set fin=1;

open old_csr;

open new_csr;

ss:loop

fetch old_csr into old_roll,old_name;

fetch new_csr into new_roll;

if fin=1 then

leave ss;

end if;

if old_roll<>new_roll then

insert into N_RollCall values(old_roll,old_name);

end if;

end loop;

close old_csr;
```

```
close new_csr;
end //
create procedure cursor_proc_p2(in r1 int)
begin
declare r2 int;
declare exit_loop boolean;
declare c1 cursor for select roll_no from O_RollCall
where roll_no>r1;
declare continue handler for not found set
exit_loop=true;
open c1;
e_loop:loop
fetch c1 into r2;
if not exists(select * from N_RollCall where roll_no=r2)
then
insert into N_RollCall select * from O_RollCall where roll_no=r2;
end if;
if exit_loop
then
close c1;
leave e_loop;
end if;
end loop e_loop;
end;//
call cursor_proc_p2(5); //
```

```
select * from O_RollCall; //
```

	roll_no	name
▶	1	Himanshu
	2	Ram
	3	Soham
	5	Mohan
	6	Om
	9	Yash
	11	Mayur

```
select * from N_RollCall; //
```

	roll_no	name
▶	6	Om
	9	Yash
	11	Mayur

```
call cursor_proc_p2(3); //
```

```
call cursor_proc_p1(); //
```

```
select * from O_RollCall; //
```

	roll_no	name
▶	1	Himanshu
	2	Ram
	3	Soham
	5	Mohan
	6	Om
	9	Yash
	11	Mayur

```
select * from N_RollCall; //
```

	roll_no	name
▶	6	Om
	9	Yash
	11	Mayur
	5	Mohan
	1	Himanshu
	2	Ram
	3	Soham

PRACTICAL – 8

```
create database lib;
```

```
use lib;
```

```
create table library(bno int(5),bname varchar(40),author varchar(20),allowed_days int(5));
```

```
create table library_audit(bno int(5),old_all_days int(5),new_all_days int(5));+.
```

```
insert into library values(1,'Database Systems','Connally T',10);
```

```
insert into library values(2,'System Programming','John Donovan',20);
```

```
insert into library values(3,'Computer Network & Internet','Douglas E. Comer',18);
```

```
insert into library values(4,'Agile Project Management','Ken Schwaber',24);
```

```
insert into library values(5,'Python for Data Analysis','Wes McKinney',12);
```

```
select * from library;
```

	bno	bname	author	allowed_days
▶	1	Database Systems	Connally T	10
	2	System Programming	John Donovan	20
	3	Computer Network & Internet	Douglas E. Comer	18
	4	Agile Project Management	Ken Schwaber	24
	5	Python for Data Analysis	Wes McKinney	12

```
delimiter //
```

```
create trigger tr1
```

```
before update on library
```

```
for each row
```

```
begin
```

```
insert into library_audit values(new.bno,old.allowed_days,new.allowed_days);
```

```
end //
```

```
update library set allowed_days=15 where bno=1; //
```

```
update library set allowed_days=25 where bno=2; //
```

```
update library set allowed_days=13 where bno=3; //
```

```
update library set allowed_days=19 where bno=4; //
```

```
update library set allowed_days=17 where bno=5; //
```

```
update library set allowed day=18 where bno=6; //
```

```
select * from library; //
```

	bno	bname	author	allowed_days
▶	1	Database Systems	Connally T	15
	2	System Programming	John Donovan	25
	3	Computer Network & Internet	Douglas E. Comer	13
	4	Agile Project Management	Ken Schwaber	19
	5	Python for Data Analysis	Wes McKinney	17

```
select * from library_audit; //
```

	bno	old_all_days	new_all_days
▶	1	10	15
	2	20	25
	3	18	13
	4	24	19
	5	12	17